

2. Objectives

The objective of this task is to develop a practical **Shopping Bill Generator using Python and Streamlit**. The application allows users to enter product names, quantities, and prices and automatically calculates the subtotal, discount, tax, and final payable amount. This task helps strengthen knowledge of Python lists, dictionaries, loops, functions, arithmetic operations, input validation, and formatted output while applying them to a real-world billing scenario.

3. Key Features

The application provides a simple interface for entering multiple products and their corresponding quantities and prices. It calculates the total price for each product and combines them to determine the subtotal. Users can also specify a discount percentage and tax/GST percentage, after which the application calculates the discount amount, taxable amount, tax amount, and final bill amount.

Main Features

- Multiple product entries.
- Product name input.
- Quantity input.
- Price input.
- Automatic item-wise total calculation.
- Formatted shopping bill.
- Discount calculation.
- Tax/GST calculation.
- Final payable amount.
- Basic input validation.
- Streamlit local-host interface.

4. Technologies Used

Python is used as the main programming language for implementing the billing logic and calculations. **Streamlit** is used to create the interactive web interface and display the generated bill. Python lists and dictionaries are used to store product information, while loops and arithmetic operations are used to process the billing data.

Technologies

- Python
- Streamlit
- Lists
- Dictionaries
- Loops
- Functions
- Conditional statements
- Arithmetic operations
- Input validation

5. Deliverables

- Shopping bill generator program.
- Itemized product bill.
- Subtotal calculation.
- Discount calculation.
- Tax/GST calculation.
- Final amount calculation.
- Streamlit-based user interface.
- Sample bill output.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY15>

```
import streamlit as st
st.set_page_config(
    page_title="Shopping Bill Generator",
    page_icon="🛒",
    layout="centered"
)

st.title("🛒 Shopping Bill Generator")
st.write("Enter product details to generate a formatted shopping bill.")

st.subheader("Add Products")

num_products = st.number_input(
    "Number of Products",
    min_value=1,
    max_value=20,
    value=3,
    step=1
)

products = []

for i in range(int(num_products)):
    st.markdown(f"### Product {i + 1}")

    col1, col2, col3 = st.columns(3)

    with col1:
        product_name = st.text_input(
            "Product Name",
            key=f"name_{i}"
        )

    with col2:
        quantity = st.number_input(
            "Quantity",
            min_value=1,
            value=1,
            step=1,
            key=f"quantity_{i}"
        )

    with col3:
        price = st.number_input(
            "Price (₹)",
            min_value=0.0,
            value=0.0,
```

```
            step=0.01,
            key=f"price_{i}"
        )

    products.append({
        "name": product_name,
        "quantity": quantity,
        "price": price
    })

st.subheader("Bill Settings")

col1, col2 = st.columns(2)

with col1:
    discount_percent = st.number_input(
        "Discount (%)",
        min_value=0.0,
        max_value=100.0,
        value=0.0,
        step=1.0
    )

with col2:
    tax_percent = st.number_input(
        "Tax / GST (%)",
        min_value=0.0,
        max_value=100.0,
        value=18.0,
        step=1.0
    )

if st.button("📄 Generate Bill", use_container_width=True):

    valid_products = [
        product for product in products
        if product["name"].strip() and product["quantity"] > 0
    ]

    if not valid_products:
        st.error("Please enter at least one valid product.")
    else:

        subtotal = 0

        for product in valid_products:
            product["total"] = product["quantity"] * product["price"]
```

OUTPUT:_http://localhost:8501

 **Shopping Bill Generator**

Enter product details to generate a formatted shopping bill.

Add Products

Number of Products:

Product 1

Product Name: Quantity: Price (₹):

Product 2

Product Name: Quantity: Price (₹):

Product 3

Product Name: Quantity: Price (₹):

Product 4

Product Name: Quantity: Price (₹):

Product 5

Product Name: Quantity: Price (₹):

Product 6

Product Name: Quantity: Price (₹):

Product 7

Product Name: Quantity: Price (₹):

Product 8

Product Name: Quantity: Price (₹):

Bill Settings

Discount (%) Tax / GST (%)

Bill generated successfully!

Shopping Bill

Product Details

crème — 2 × ₹20.00 = ₹40.00

Powder — 3 × ₹90.00 = ₹270.00

Grocery — 3 × ₹100.00 = ₹300.00

Paneer — 2 × ₹180.00 = ₹360.00

Ice cream — 2 × ₹90.00 = ₹180.00

lays — 5 × ₹50.00 = ₹250.00

vegetables — 3 × ₹120.00 = ₹360.00

books — 4 × ₹180.00 = ₹720.00

Subtotal ₹2480.00

Discount • ₹248.00

Amount After Discount ₹2232.00

Tax / GST • ₹379.44

Final Amount ₹2611.44

Thank you for shopping! 🛒

Github:- <https://github.com/shambhavi0804/VEDA-internship/commit/f0022ee4df17ed5896a35799d3c40a5d1fa8f5a3>

```
DAY15/app.py
*** @ -> 0,0 +1,140 @
1 + import streamlit as st
2 + st.set_page_config(
3 +     page_title="Shopping Bill Generator",
4 +     page_icon="🛒",
5 +     layout="centered"
6 + )
7 +
8 + st.title("🛒 Shopping Bill Generator")
9 + st.write("Enter product details to generate a formatted shopping bill.")
10 +
11 + st.subheader("Add Products")
12 +
13 + num_products = st.number_input(
14 +     "Number of Products",
15 +     min_value=1,
16 +     max_value=20,
17 +     value=1,
18 +     step=1
19 + )
20 +
21 + products = []
22 +
23 + for i in range(int(num_products)):
24 +     st.markdown(f"### Product {i + 1}")
25 +
26 +     col1, col2, col3 = st.columns(3)
27 +
28 +     with col1:
29 +         product_name = st.text_input(
30 +             "Product Name",
31 +             key=f"name_{i}"
32 +         )
33 +
34 +     with col2:
```

Linkedin:- <https://lnkd.in/p/dVNDASqt>

Shopping Bill

Product Details

****crème**** — $2 \times ₹20.00 = ₹40.00$

Powder — $3 \times ₹90.00 = ₹270.00$

Grocery — $3 \times ₹100.00 = ₹300.00$

****Paneer**** — $2 \times ₹180.00 = ₹360.00$

****ice cream**** — $2 \times ₹90.00 = ₹180.00$

lays — $5 \times ₹50.00 = ₹250.00$

****vegetables**** — $3 \times ₹120.00 = ₹360.00$

books — $4 \times ₹180.00 = ₹720.00$

Subtotal	₹2480.00
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Discount • ₹248.00

Amount After Discount	₹2232.00
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Tax / GST	• ₹379.44
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Final Amount ₹2611.44

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